

## 7 Prototypes, 2 Years, 1 Patent: How an IIT Kanpur Student Reinvented the Shoe

### *Why Schault Decided the Whole Shoe Never Needed to Be Thrown Away*

Meta Title: 7 Prototypes, 2 Years, 1 Patent — How Schault Reinvented the Shoe

Meta Description: An inside look at how Schault, an IIT Kanpur startup, built seven prototypes over two years to create a patented modular footwear system that lets users replace worn-out soles and uppers instead of discarding the entire shoe.

When did you last wash your shoes? Not wipe them down, actually wash them. If you're hesitating, you're not alone: nearly 60% of users report avoiding a pair of shoes altogether because it felt unclean. And here's the harder question and why did you throw away the whole shoe when only the sole had worn out?

That single, deceptively simple question sits at the center of this story. Every year, roughly 22 billion pairs of shoes are discarded worldwide, and more than 90% of them end up in landfills not because the entire shoe failed, but because one part did. This blog walks through how a student at IIT Kanpur turned that frustration into a patented, modular footwear system through seven prototypes, dozens of failures, and two years of relentless iteration. If you're building a hardware startup, or you're just curious what real product development looks like behind the scenes, this is that story.

## The Problem: Footwear Is Broken Across Three Dimensions

Before any prototype existed, the Schault team mapped out exactly why footwear, as a category, was failing consumers:

- **Hygiene Crisis:** Sweat, dirt, and odour get trapped inside shoes, causing real social anxiety around 60% of users report hygiene concerns with their footwear.
- **Cost Inefficiency:** An entire pair gets replaced when only the sole or upper fails, wasting upwards of ₹3,000 per discarded pair.
- **Waste Catastrophe:** With 22 billion pairs discarded yearly and multi-material composites that are nearly impossible to recycle, 90–95% of footwear ends up landfilled or incinerated. The irony, as the founders point out, is that India is the world's second-largest shoe manufacturer yet the category has seen almost no real innovation in decades.



## Prototype 1: Early Experiments with Cardboard and Nails



Every hardware product and yes, a shoe is a hardware product starts with the cheapest possible way to test an idea. Our very first prototype wasn't glamorous: a basic canvas sneaker with a sole crudely attached using cardboard as a structural spacer and nails as fasteners.

The goal at this stage was never comfort, aesthetics, or durability. It was singular: \*prove that our core sole-attachment concept could physically hold together. Before spending money on real rubber compounds, custom tooling, or manufacturing partnerships, we needed to know the fundamental idea had legs (pun intended).

Cardboard was used to simulate the rigid mid-layer that would eventually separate the upper from the outsole, while nails served as a stand-in for a future mechanical fastening system.

Why it matters for stakeholders: This "kitchen-table prototyping" stage reflects a lean, capital-efficient R&D philosophy. Instead of burning capital on tooling for an unvalidated idea, we spent under a dollar to get our first directional signal. This is the same build-test-learn discipline seen in successful hardware startups across consumer electronics and industrial design.

## Prototype 2: Initial Sole Prototype — Interlocking Failure



With the core concept validated, we moved to our first dedicated sole prototype and a standalone rubber outsole engineered with ridged, grooved patterns. The intent was to create a mechanical interlock between the sole and the shoe's upper, removing dependency on adhesives or stitching, which are common failure points in traditional shoe construction (delamination, glue degradation, seam splitting).

Upon testing, the interlock did not hold. The ridges and grooves failed to lock into a stable position under the stresses of normal wear, and the sole risked separating from the upper.

Root cause analysis: This failure pointed to two probable issues

1. Insufficiently precise tolerances between the interlocking features
2. Rubber's inherent flexibility, which made a purely rubber-based mold difficult to keep dimensionally stable during the interlock process.

Why it matters: Failure at this stage was valuable, not wasteful. It told us precisely where the mechanical design needed rework as, namely, tighter tolerance control and possibly a more rigid material for the interlocking teeth or channels. This is a textbook example of "failing fast" in physical product development.

## Prototype 3: First MVP with Functional Challenges



Armed with lessons from Prototype 2, we built our first genuinely wearable Minimum Viable Product: a mesh slip-on shoe with an integrated sole. This was a significant leap for the first time, someone could actually put the shoe on and walk in it

However, real-world wear testing surfaced multiple functional issues. These likely included:

- Fit inconsistency - The sole-to-upper relationship didn't sit uniformly across different foot shapes or sizes.
- Bonding weakness - The connection between sole and upper wasn't strong enough for repeated flexing during walking.
- Durability concerns - Signs of wear appeared faster than acceptable for a commercial product.

Why it matters: This stage answered the most important early question in footwear development "can this actually be worn?" with a qualified yes. It also gave us a clear punch list of engineering problems to solve before we could responsibly move toward manufacturing at any scale.

## Prototype 4: First 3D-Printed Prototype — Interlocking Issues



To break out of the precision limitations of hand-cast rubber, we brought in 3D printing and a pivotal shift in our R&D methodology. 3D printing lets us iterate on the interlocking pin, groove, and hole geometry with far greater dimensional accuracy, and at a fraction of the cost and time of traditional mold-based tooling.

Despite the improved manufacturing precision, the interlocking pins and knobs still failed to seat correctly into the corresponding channels on the upper. This is a common and expected hurdle when translating a digital CAD design into a physical, load-bearing mechanical fit; small tolerance mismatches that look fine on-screen often reveal themselves only in physical testing.

Why it matters: This stage dramatically accelerated our development velocity. What might have taken weeks per iteration using traditional mold-making now took days using 3D printing and letting us test more design variations, faster, and at lower cost. It also proved out 3D printing as a viable rapid-prototyping tool for our future sole R&D.

## Prototype 5: Refined 3D-Printed Prototype



Using precise data from the Prototype 4 failure, we refined the pin, knob, and channel geometry to achieve a tighter, more secure mechanical interlock. This wasn't a redesign from scratch, it was a targeted, data-driven refinement of specific dimensions and structural features that had caused the previous misalignment.

Why it matters: This is the stage where our proprietary interlocking sole attachment system matured from a promising idea into a validated, repeatable engineering solution. It represents the resolution of the single biggest technical risk in the entire product roadmap reliably joining sole to upper without glue or stitching.

## Prototype 6: Final MVP Development



With the core interlock mechanism now proven, we shifted focus from engineering validation to product realization and translating the refined design into premium, real-world materials. This MVP used genuine leather uppers in tan and cream tones, paired with the now-validated sole architecture from Prototype 5.

Why it matters: This stage represents the transition from "engineering test rig" to genuine consumer product. It's the first version of the shoe that looks, feels, and performs like something that could actually sit on a retail shelf while still relying on the same interlocking sole technology validated in earlier, less polished stages.

## Prototype 7: The Final PU-Cast Prototype



In our final and most production-critical stage, we transitioned from 3D-printed soles to \*PU (Polyurethane) casting\* — the industry-standard process used by major global footwear manufacturers for producing durable, lightweight, flexible soles at scale.

This transition matters enormously from a manufacturing standpoint. 3D printing is an outstanding tool for \*prototyping\* — fast, cheap, and flexible for iteration and but it is not a viable process for \*mass production\* due to cost-per-unit and throughput limitations. PU casting, by contrast, is purpose-built for high-volume, consistent, cost-effective manufacturing.

PU casting involves injecting liquid polyurethane into a precision mold, allowing it to cure into a durable, flexible sole with consistent density and cushioning properties — far superior to 3D-printed materials in terms of long-term wear resistance and comfort.

Why it matters: This final stage is the clearest signal to investors and manufacturing partners that the product is no longer a prototype — it is \*production-ready.\* The core technical bottleneck (sole-to-upper interlock) has been solved, refined, and now translated into an industry-standard, scalable manufacturing process.

## The Business Case: Why Modularity Also Makes Financial Sense

- Beyond the sustainability story, Schault's model is built on razor-and-blade economics and a base shoe followed by recurring, high-margin part replacements:
- Base shoe: ₹3,000 selling price, ₹1,500 manufacturing cost, ~50% gross margin.
- Replacement upper: ₹2,000 — Replacement sole: ₹1,000, both largely recurring revenue.
- Roughly 50% of customers are expected to buy at least one replacement component, effectively doubling customer lifetime value.

- Projected 3-year revenue: ₹1.50 crore across 5,000 pairs sold, with an LTV/CAC ratio above 3x.

This is the founder-level insight that matters most for other hardware entrepreneurs: sustainability and recurring revenue aren't in tension in the Schault's model, they're the same design decision.

## What Seven Prototypes Taught the Founders

- Fail fast: the faster you test an idea, the faster you learn whether it's worth pursuing.
- Build in public: sharing the journey openly brought feedback and validation the team couldn't have generated alone.
- Iteration beats perfection: none of the early prototypes were perfect progress came from continuous, incremental improvement.
- Sustainability requires real innovation: reducing waste isn't only about eco-friendly materials — sometimes the product itself has to be redesigned.

### VISUALS

The full seven-stage evolution, side by side:



*From cardboard and nails to a patented, production-ready modular footwear system — the complete Schault prototype journey.*

### CONCLUSION

Seven prototypes. Dozens of failures. Hundreds of design revisions. Countless hours of testing. What began as an experiment with cardboard and nails has evolved into a patented modular footwear system designed for a more sustainable future and validated with a manufacturing partner already onboarded.

*The bigger takeaway isn't really about shoes. It's a reminder that most meaningful hardware innovation looks like failure right up until it doesn't. If you're building something physical, ask yourself: which part of your product are people quietly tolerating instead of replacing and what would it take to let them fix just that part?*



#### CALL-TO-ACTION

**Schault is currently raising a ₹1 crore pre-seed round to launch its D2C website, hit its first 2,000 pairs sold, and validate an LTV/CAC ratio above 3x within 12–15 months.**

- Interested in the opportunity? Reach out to Harsh Maheshwari (Founder & CEO) at [harshm23@iitk.ac.in](mailto:harshm23@iitk.ac.in) or +91 8755054200.
- Explore the product: [schault.com](https://schault.com)
- Follow the journey and share your thoughts every piece of feedback shapes the next prototype.